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Prospective remembering of Korsakoffs and alcoholics as a function of the prospective-memory and on-going tasks

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Abstract

Prospective memory is assumed to rely more on the frontal lobes than retrospective memory. Since Korsakoff patients are known to suffer from a general cerebral atrophy and a frontal lobe atrophy in particular, they are expected to show considerably impaired prospective memory. In Experiment 1, the performance of Korsakoff patients on a semantic prospective-memory task (which was embedded in a perceptual on-going task) was particularly bad in Session 1; in Session 2, the Korsakoff patients improved substantially, to reach the performance level of nonamnesic alcoholics. In Experiment 2, prospective memory of the Korsakoff patients and nonamnesic alcoholics was better when the on-going task was more similar to the prospective-memory task; particularly striking was the much better prospective memory in the semantic prospective-memory task when the on-going task requires a semantic analysis than when the on-going task requires perceptual processing. The findings are in agreement with a task-appropriate processing explanation but also in partial agreement with the attention hypothesis of the instance theory of automaticity. Contrary to the frontal lobe hypothesis, prospective memory of the Korsakoff patients was surprisingly good in several aspects of the two experiments. © 2000 Elsevier Science Ltd. All rights reserved.

Keywords: Prospective memory; Korsakoff; Alcoholics

1. Introduction

An easy way to describe prospective memory is to contrast it with the more familiar retrospective memory. Retrospective remembering stands for remembering information about activities that happened in the past. Prospective remembering, on the other hand, involves remembering activities to be performed in the future, as for example remembering to call a friend or to post a letter. Two forms of prospective memory have been distinguished [14,15]. Event-based prospective remembering stands for remembering to perform an action when some particular external event (called the target event) occurs; for example, asking a question to a friend when you meet him or her. Time-based prospective remembering means to remember to perform an action at a certain point in time (called the target time); for example, calling a friend at noon today. Time-based prospective remembering is assumed to be more difficult because it requires more self-initiated retrieval processes than event-based prospective remembering [15].

The two forms of memory, prospective and retrospective memory, are not totally independent. A prospective-memory task typically also involves a retrospective component, namely the 'when' and 'what' of the task. For example, remembering to call a friend when you get home, requires also to 'when' to do this (when you get home) and 'what' to do then (to call). Many prospective-memory failures probably occur because of problems in remembering the retrospective component [14].

Research on the underlying brain structures suggests

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that prospective functioning relies more on the frontal lobes than retrospective memory [3,18,20,35]. The decision about what action needs to be performed, and the creation of the intent to carry out fall within the traditional domain of the executive function [5]. The frontal lobes are responsible for formulating plans, initiating actions, monitoring on-going behavior, and evaluating outcomes, and therefore probably also for prospective remembering [19]. When cognitive deficits appear in frontal lobe dementias, they typically involve disorders of planning and organization [38].

Korsakoff patients are known to have serious difficulties with retrospective memory. They suffer from severe anterograde and retrograde amnesia [6–8,24]. Far less is known about their prospective-memory functioning. While the precise anatomical localization of the Korsakoff syndrome remains a problem, it is assumed that there is damage in the medial thalamus, and possibly in the mammillary bodies of the hypothalamus as well as a generalized cerebral atrophy especially in the frontal lobe [21,23,36,37]. A prospective-memory task always implies retrospective- and prospective-memory components. The amnesic nature of the Korsakoff patients will greatly affect the retrospective component. Their frontal lobe atrophy will additionally influence the prospective component. They do show impairment in the planning of behavior and the formation of judgment [17,37]. Therefore, we predict that Korsakoff patients will show a considerably impaired prospective memory.

In a pilot study with Korsakoff patients [9], time- and event-based prospective memory was studied, using the identification of famous faces on slides as an on-going task. The pilot study was a failure as most Korsakoff patients refused to continue the task. It became clear that the tasks need to be easy in order to get the Korsakoff patients sufficiently involved. In the present study, the following changes were made. As a time-based prospective-memory task is assumed to be more demanding, only an event-based prospective-memory task was used. Identifying faces involves retrospective memory; therefore, another on-going task was selected. Since Korsakoff patients do not have problems with perceptual tasks [4], counting letters in a word seemed a better on-going task. As Coeckelbergh [9] observed a considerable improvement between a first and second session (with only a few minutes break in-between), we also used two sessions.

We also took into account the training some Korsakoff patients received to overcome their memory problems. In some psychiatric institutions, Korsakoff patients are given a so-called diary training. The purpose is to train Korsakoff patients in using a diary as a kind of memory reminder for the daily activities. The availability of the reminder was manipulated in order to see whether the reminder was indeed more useful for trained than untrained Korsakoff patients, in carrying out the prospective-memory task. As a control nonamnesic group, chronic alcoholics were selected who resided in similar institutions as the Korsakoff patients and who showed no memory impairments on a standard memory test.

2. Experiment 1

2.1. Method

2.1.1. Participants

Ten trained and 10 untrained Korsakoff patients, and 10 alcoholics participated in Experiment 1. All 10 (one female and nine male) trained Korsakoff patients were residents in a psychiatric institution. The average age of the group was 49.1 years (range 42–59 years). All patients showed severe anterograde and retrograde amnesia. On the Auditory–Verbal Learning Test, all scores were in the severe memory deficit range. For five of those patients we had scores on the Wechsler Adult Intelligence Scale, with an average VIQ of 84.8, an average PIQ of 89.8, and an average overall IQ of 88.2. For four other patients, the average Progressive Matrices IQ was 91. For one patient there was no IQ available.

The untrained Korsakoff group consisted of two female and eight male patients from two different psychiatric institutions. The average age of the group was 52.2 years (range 48–59 years). All patients also showed severe anterograde and retrograde amnesia. On the Auditory–Verbal Learning Test, the scores were all in the severe memory deficit range. For four patients we had an IQ approximation by the scores on the ‘Nederlandse Leeslijst voor Volwassenen’ [Dutch Reading List for Adults], with an average IQ of 98.75. For the remaining six patients, the average Wechsler Adult Intelligence Scale IQ was 86.1; there were no separate VIQ and PIQ scores available.

The 10 male chronic alcoholics were hospitalized in a detoxication ward of a similar psychiatric institution. The average age was 41.8 years (range 35–59 years). Among the alcoholics, there were no signs of any memory problems. All scores on the Auditory–Verbal Learning Test were situated in the no-memory deficit range. They were given no medication that could affect memory. The average IQ on the Progressive Matrices was 108.1.

All 30 participants had abstained from alcohol for at least one month.

2.1.2. Design

Participants were tested twice individually. In Session 1, each participant was randomly assigned to one of the two conditions (reminder vs no reminder). In

Session 2 which followed Session 1 after a 1-min break, the participant was assigned to the same experimental condition. This results in a 3 × 2 × 2 design, with participant group (trained and untrained Korsakoff patients, and alcoholics) and availability of reminder (reminder vs no reminder) as between-subjects variables and session (Session 1 vs Session 2) as a within-subjects variable.

2.1.3. Materials

Each session involved the presentation of a different series of 22 nouns. Each word was presented for as long as the participants needed. When participants had finished counting the letters of a word, a next word appeared. In each series there were eight target words, defined as words that belong to the category of animals. In Session 1 the target words were located on Positions 3, 5, 8, 10, 14, 17, 20, and 22. In Session 2 the target words were located on positions 2, 7, 9, 12, 14, 17, 19, and 21.

A computer screen was placed in front of the participants at a distance of approximately 40 cm. The space bar of the keyboard was colored red. On the right of the keyboard there was a leaflet with the printed instructions. In the reminder conditions, there was a blank leaflet on the left of the keyboard.

2.1.4. Procedure

Entering the room, the experimenter introduced herself and thanked the participant for the cooperation. Participants could follow the printed instructions while the experimenter was reading them aloud. No information was given about the aim of the experiment. The participants were told that words were to be shown on the screen. They were asked to count the letters of the words, and to enter the number on the keyboard. They were told that by entering the number, the next word would be presented. A second instruction was given, referring to the prospective-memory task component. Words that belong to the category of animal were told to be 'special': no counting was required for these words. Instead, participants had to push the red space bar of the keyboard.

Before the experiment started, participants in the reminder condition were instructed to write on a leaflet the retrospective-memory components of the task: the 'when' component (animal) and the what component (the red button). In the reminder condition of the trained Korsakoff group, this leaflet was replaced by their own diary. No records could be taken how often the participants looked at the leaflet or diary during the task. No leaflet or diary was given in the no-reminder conditions.

After the instructions were given, participants had the opportunity to ask questions. When there were no further questions, the first word was presented. When participants had finished all words of Session 1, a 1-min break was given with the instructions before Session 1 being repeated.

2.2. Results

2.2.1. Letter counting task

The analysis of variance involves the three groups of participants and the availability of a reminder as between-subjects variables, and session as a within-subjects variable. The proportion of words with correctly counted letters is the dependent variable. The analysis reveals no significant main and interaction effects. The mean level of performance is 0.957 and 0.954 for the trained and untrained Korsakoff patients respectively, and 0.904 for the alcoholics.

2.2.2. Prospective-memory performance

The analysis of variance includes the three groups of

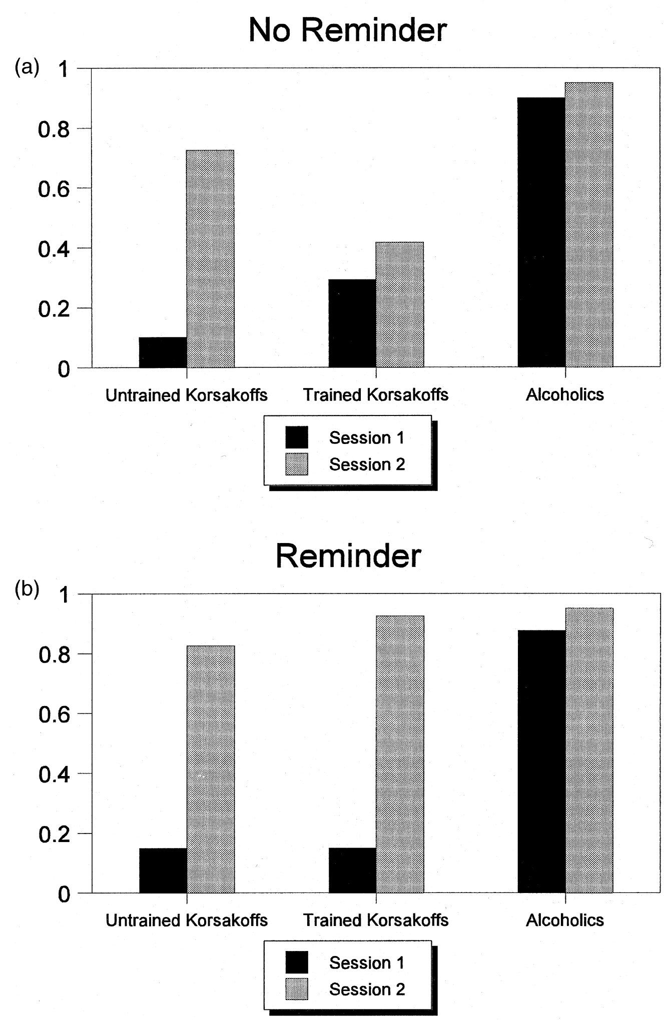

Fig. 1. Proportion of prospective-memory responses as a function of groups of participants, availability of the reminder, and sessions (Experiment 1).

participants and the availability of a reminder as between-subjects variables, and session and the eight target words as a within-subjects variables; the dependent variable is the proportion prospective-memory responses. We immediately noticed that two trained Korsakoff patients in the no-reminder condition showed a deviant score in Session 1: while the proportions prospective-memory responses of the remaining trained and untrained Korsakoff patients were very small in Session 1 (M = 0.22 and 0.13 respectively), those two participants showed perfect performance. Similarly to the whole group of Korsakoff patients, those two patients showed severe anterograde and retrograde amnesia; their scores on the Auditory–Verbal Learning Test were in the severe memory deficit range, and their overall IQ was 83 and 108, with age 48 and 57, respectively. Thus, from the available information, they did not differ from the other Korsakoff patients. We therefore decided to carry out the analysis twice, one with and the other without the scores of those two participants. We first present the outcome of the analysis without those two participants; Fig. 1 is based on the data without those two participants. Thereafter, we describe the (small) difference in findings when the data from the two participants are included.

There is a significant main effect of the groups of participants, $F(2,22) = 11.276$, $MSE = 1.000$, $P < 0.0005$. Following a posteriori Tukey tests, it appears that Korsakoff patients (both trained and untrained) perform significantly (at least $P < 0.05$) worse (M = 0.446 and M = 0.450 respectively) than alcoholics (M = 0.919). The difference between the trained and untrained Korsakoff patients is not significant. There is a significant main effect of session, $F(1,22) = 45.887$, $MSE = 0.353$, $P < 0.00001$, with $M = 0.411$ in Session 1 and with $M = 0.799$ in Session 2. The order of the eight target words also produces a significant main effect, $F(7,154) = 2.237$, $MSE = 0.053$, $P < 0.04$, due to a significant linear increase of performance from Target Word 1 ($M = 0.533$) to Target Word 8 ($M = 0.622$), $F(1,22) = 4.765$, $MSE = 0.110$, $P < 0.04$.

There is a significant interaction effect involving the three groups of participants and the two sessions, $F(2,22) = 10.038$, $MSE = 0.353$, $P < 0.001$. A clear pattern of findings is apparent (see Fig. 1). There is an important improvement of performance from Session 1 to Session 2 among the two groups of Korsakoff patients (M = 0.125 and 0.775 for untrained, and M = 0.221 and 0.671 for trained Korsakoff patients; in both cases, $P < 0.05$ with a Tukey test), while the alcoholics perform almost at ceiling level at the two sessions (M = 0.888 and 0.950; the difference is not significant). In Session 1, the alcoholics perform significantly better than the two Korsakoff groups (Tukey test, $P < 0.05$), while all three groups perform about at the same level in Session 2.

Another significant first-order interaction effect includes the two sessions and the order of the eight target words, $F(7,154) = 2.763$, $MSE = 0.052$, $P < 0.01$. While there is a significant linear increase from Target Word 1 ($M = 0.256$) to Target Word 8 ($M = 0.489$) at Session 1 [$F(1,22) = 6.789$, $MSE = 0.828$, $P < 0.02$], no difference appears at Session 2 (overall $M = 0.800$).

A third significant first-order interaction effect, availability of reminder by order of target words, $F(7,154) = 2.787$, $MSE = 0.053$, $P < 0.01$, is involved in a significant second-order interaction effect which includes the effect of the three groups of participants, $F(14,154) = 1.914$, $MSE = 0.053$, $P < 0.03$. This second-order interaction is mainly due to a strong linear increase of performance among alcoholics as a function of target position in the reminder condition, $F(1,22) = 9.057$, $MSE = 0.110$, $P < 0.007$, while the same trend analysis in the other conditions fails to produce significant effects.

The availability of reminders interacts significantly with the two sessions, $F(1,22) = 4.462$, $MSE = 0.353$, $P < 0.05$. At Session 1, the availability of reminders does not affect performance (M = 0.431 and 0.392, without and with reminder respectively) while at Session 2, performance is better with reminder than without reminder (M = 0.900 and 0.697; marginally significant following a Tukey test, $P < 0.08$). This difference is almost entirely due to a much smaller increase from Session 1 to Session 2 among trained Korsakoff patients without a reminder, as can be seen in Fig. 1, producing a marginally significant interaction between the three groups of patients, availability of reminder, and session, $F(2,22) = 2.900$, $MSE = 0.353$, $P < 0.08$.

When the analysis is repeated including this time the two trained Korsakoff patients who showed perfect prospective-memory performance in Session 1, the same significant effects are obtained; this time however, the interaction between the group of participants, the availability of the reminder and the two sessions becomes significant, $F(2,24) = 4.478$, $MSE = 0.327$, $P < 0.03$.

Adding age, IQ (whenever available) or the scores on the Auditory–Verbal Learning Test as covariates in the analyses of variance does not change the pattern of findings. Moreover, interactions of the covariate age, IQ or the Auditory–Verbal Learning Test with group or availability of reminder are not significant. There are almost zero correlations between age, IQ, or Auditory–Verbal Learning Test, and the proportion prospective-memory responses (pooled over Session 1 and 2, and partialling out the effects of the groups).

2.3. Discussion

The most salient findings from Experiment 1 is the poor prospective-memory performance of the two Korsakoff groups at Session 1. Except for the two Korsakoff patients with perfect performance, the performance of the remaining Korsakoff patients is extremely poor, 0.22 and 0.13 for the trained and untrained groups respectively. One could conjecture that the participants either responded to all of the target words or to virtually none of the target words. This is consistent with a memory explanation of the findings, namely that most Korsakoff patients forgot what they were supposed to do, but when reminded at the end of Session 1, they then performed almost normally in Session 2. This conjecture is not likely as the three groups of participants showed a linear increase of performance from Target Word 1 to 8 in Session 1.

There is one exception to the improvement among Korsakoff patients from Session 1 to Session 2: trained Korsakoff patients who were not entitled to use their dairy (no reminder condition) did not improve significantly. Perhaps the absence of a diary negatively affected their performance at Session 2 as they are used to work with a diary. However, we expected the Korsakoff patients who were trained to use a diary to show a better performance when they were entitled to use their diary during the experiment; this did not happen: basically, there is no difference between the trained and untrained Korsakoff patients when the reminder was available (see Fig. 1b). In fact, there is no direct evidence in the data suggesting that the reminder was used to perform the prospective-memory task. Furthermore, although no systematic records were taken how often the participants checked the leaflet or diary in the reminder conditions, casual observations by the experimenter suggest that the leaflet or diary was never used.

Clearly, the presence of an external reminder did not prompt the Korsakoff patients to perform the prospective-memory task more successfully. Perhaps they were too involved in performing the on-going task which was very different. The on-going task of counting letters in a word and the prospective-memory task of pushing a button when a word belongs to the category of animal in Experiment 1 require indeed different cognitive processes: counting letters requires words to be processed at a more superficial, perceptual level, and reacting to words that belong to a semantic category requires that the words are processed at a deeper, more semantic level. In Experiment 2, the nature of the on-going and prospective-memory tasks was manipulated. The on-going task was either perceptual or semantic; similarly, the prospective-memory task required either perceptual or semantic processing. Accordingly, the on-going task in some conditions was made more similar to the prospective-memory task (i.e., both tasks requiring either perceptual or semantic processing) such that some responses to the on-going task are prompting the participants to the second, prospective-memory task. According to the task-appropriate processing hypothesis (which will be presented in the General Discussion), the similarity in processing may facilitate prospective-memory performance.

3. Experiment 2

3.1. Method

3.1.1. Participants

Two new samples of participants were tested: 24 male and three female untrained Korsakoff patients and 24 (20 men and four women) alcoholics. All participants had abstained from alcohol for at least one month. The 27 Korsakoff patients were residents in a psychiatric institution. The average age of the group was 50.6 years (range 37–64 years). All patients showed severe anterograde and retrograde amnesia. On the Auditory–Verbal Learning Test, all scores were in the severe memory range. Their general cognitive functioning however was not affected. On the Wechsler Adult Intelligence Scale (WAIS), the average overall IQ was 101.9.

The chronic alcoholics were hospitalized in a detoxication ward of a psychiatric institution. The average age was 41.8 years (range 35–59 years). Among the alcoholics, there were no signs of any memory problems. All scores on the Auditory–Verbal Learning Test were situated in the no-memory deficit range. The average overall IQ (WAIS) was 108.1.

3.1.2. Design

All participants were tested twice individually. Half of the participants received in the two sessions the perceptual prospective-memory task, while the other participants received the semantic prospective-memory task. In Session 1, each participant was randomly assigned to one of the four further experimental conditions (perceptual vs semantic on-going task and the availability of a reminder). In Session 2 which followed Session 1 after a 1-min break, the participant was assigned to the same experimental condition, except for receiving the other on-going task. This results in a 2 (Korsakoff patient vs alcoholic) $\times$ 2 (reminder vs no reminder) $\times$ 2 (perceptual vs semantic prospective-memory task) $\times$ 2 (first perceptual then semantic on-going task vs first semantic then perceptual on-going task) between-subjects variables.

3.1.3. Procedure

The materials and procedure of Experiment 2 were the same as in Experiment 1, except for the following. Half of the participants received in Session 1 the instruction to count the letters of the presented words and to enter this number on the keyboard in front of them (perceptual on-going task). The other participants received the instruction to explain briefly the meaning of the words (semantic on-going task) and then to push the yellow button of the keyboard to get the following word.

Participants in the semantic prospective-memory task were given the additional instruction to push the red button of the keyboard whenever a word belonged to the category of animal while participants in the perceptual prospective-memory task were given the instruction to push the red button whenever the presented word contained five letters.

After Session 1, participants who were first given the instruction to count the letters of words in Session 1 (perceptual on-going task), now received the instruction to give the meaning of the words (semantic on-going task), or vice versa. Both groups again were given the instruction to push the red button whenever a word belonged to the category of animal or contained five letters (depending on the prospective-memory condition).

Some participants performed both the on-going task and the prospective-memory task on the prospective-memory items, and this was still counted as correct on the prospective-memory task.

The experiment was carried out in two stages: first all conditions with a semantic prospective-memory task, and later (with different samples of participants), the conditions with the perceptual prospective-memory task. Unfortunately, a change was inadvertently introduced in the second stage: the manipulation of the reminder conditions with the semantic prospective-memory task in the first stage was the same as in Experiment 1; with the perceptual prospective-memory task in the second stage of the experiment, all participants had to repeat loudly the instructions of the experimenter and the participants in the reminder conditions thereafter had to write down the instructions before starting the experiment.

3.2. Results

3.2.1. Performance in on-going task

The analysis of variance on the proportion of words with correctly counted letters, as a function of the two groups of participants (Korsakoff patients and alcoholics), the two prospective-memory tasks, and the availability of the reminder shows one significant main effect, $F(1,41) = 4.175$, $MSE = 0.013$, $P < 0.05$: the proportion of correctly counted letters is larger when the prospective-memory task is perceptual ($M = 0.964$) than when it is semantic ($M = 0.894$). There was no need to conduct an analysis of variance involving the proportion correctly explained words as almost all participants displayed perfect performance ($M = 0.967$ and 1.00 for the Korsakoff patients and alcoholics).

3.2.2. Prospective-memory performance

The analysis of variance on the proportion prospective-memory responses has four between-subjects variables (Korsakoff patients vs alcoholics, the two prospective-memory tasks, the availability of the reminder, and the order of the two on-going tasks) and two within-subjects variables (the two on-going tasks and the eight target words). The analysis reveals a significant main effect of groups of participants, $F(1,33) = 17.276$, $MSE = 1.195$, $P < 0.0002$ with the alcoholics performing better ($M = 0.875$) than the Korsakoff patients ($M = 0.533$). There is also a significant main effect of type of on-going task, $F(1,33) = 5.917$, $MSE = 0.537$, $P < 0.02$. Participants

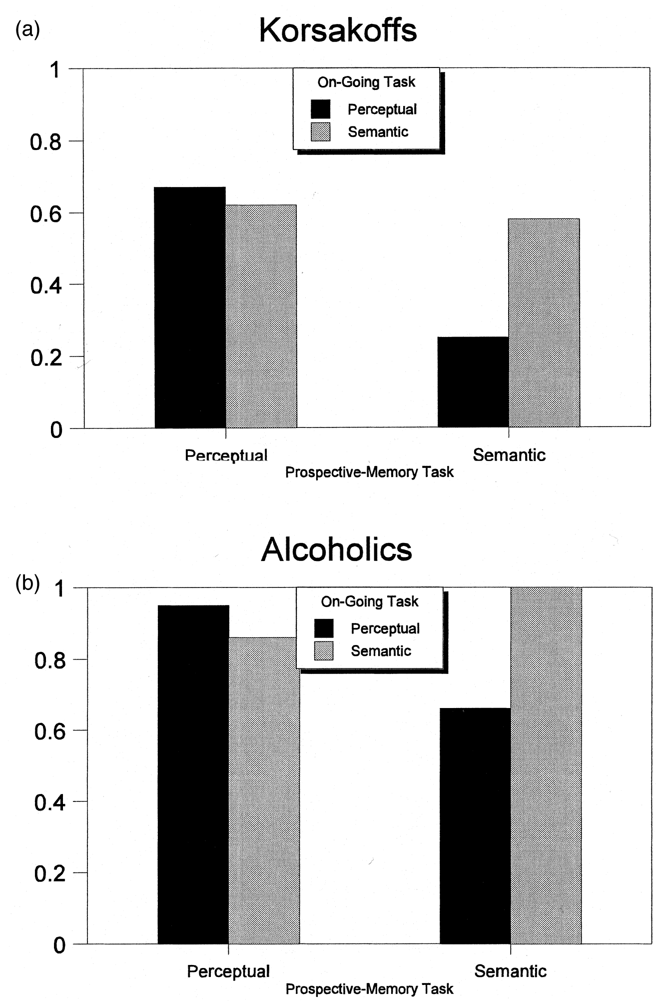

Fig. 2. Proportion of prospective-memory responses as a function of groups of participants, and the prospective-memory and on-going task (Experiment 2).

perform significantly better when the prospective-memory task is embedded in a semantic on-going task ($M = 0.771$) than when the prospective-memory task is embedded in a perceptual on-going task ($M = 0.637$). The main effect of the two prospective-memory tasks is only marginally significant, $F(1,33) = 3.479$, $MSE = 1.195$, $P < 0.07$: a perceptual prospective-memory task leads to a marginally better performance than a semantic prospective-memory task ($M = 0.781$ vs 0.628).

Critically important is the significant interaction between the nature of the prospective-memory tasks and the nature of the on-going-tasks, $F(1,33) = 12.424$, $MSE = 0.537$, $P < 0.001$, see Fig. 2. Best performance is obtained when the prospective-memory task and the on-going task are both either perceptual or semantic; particularly striking is the worst performance when the prospective-memory task is semantic and the on-going task is perceptual. A posteriori Tukey tests show that semantic prospective memory is significantly better when it is embedded in a semantic ($M = 0.792$) than in a perceptual on-going task ($M = 0.463$, $P < 0.003$) and that semantic prospective memory ($M = 0.463$) is considerably worse than perceptual prospective memory ($M = 0.811$) when both are embedded in a perceptual on-going task ($P < 0.0005$). The nature of the on-going task does not affect significantly perceptual prospective memory ($M = 0.811$ and 0.751 in a perceptual and semantic on-going task); similarly, perceptual and semantic prospective memory ($M = 0.751$ and 0.792) do not differ significantly when they are embedded in a semantic on-going task.

From Fig. 2, it is immediately apparent that the interaction is quite similar among the two groups of participants. Not surprisingly, the interaction between the two groups, the nature of the prospective-memory task, and the nature of the on-going tasks is not significant, the F value being even smaller than unity.

The analysis of variance was repeated separately on the data of Session 1 and Session 2. The interaction between the prospective-memory and on-going tasks is significant at Session 1, $F(1,33) = 5.792$, $MSE = 0.626$, $P < 0.02$, but is only marginally significant at Session 2, $F(1,33) = 2.765$, $MSE = 1.107$, $P < 0.10$. In both sessions, Korsakoff patients are significantly worse than alcoholics [Session 1: $F(1,33) = 26.341$, $MSE = 0.626$, $P < 0.00001$; and Session 2: $F(1,33) = 5.061$, $MSE = 1.107$, $P < 0.03$] but the difference is smaller in Session 2 ($M = 0.618$ and 0.870) than in Session 1 ($M = 0.449$ and 0.880) as Korsakoff patients improved their performance considerably in Session 2.

The availability of the reminders did not produce significant main or interaction effects. Similarly, the effect of the position of the eight target words as well as its interactions never reached significance.

Adding age, IQ or the scores on the Auditory–Verbal Learning Test as covariates in the analyses of variance does not change the pattern of findings. Moreover, interactions of the covariate age, IQ or the Auditory–Verbal Learning Test with group or availability of reminder are not significant. Simple correlations could not be computed with the alcoholics, as too many participants showed perfect performance. Among the Korsakoff patients, there are almost zero correlations between age, IQ, or Auditory–Verbal Learning Test, and the proportion prospective-memory responses (pooled over Sessions 1 and 2).

4. General discussion

We conjectured that the poor prospective-memory performance of the Korsakoff patients in Session 1 of Experiment 1 was due to the different nature of the two tasks, the on-going and prospective-memory task. The nature of the prospective-memory and on-going tasks in Experiment 2 was manipulated, in order to have in some conditions an on-going task sharing the same processes as the prospective-memory task; in other words, the processing requirements of the on-going task could provide the participants an internal reminder of the prospective-memory task. Indeed, prospective-memory performance was better when the on-going task was more similar to the prospective-memory task.

Critical in the present study is the nature of the prospective-memory task and its relationship to the processes in the on-going task. d'Ydewalle, Utsi, and Brunfaut [13] and d'Ydewalle, Luwel, and Brunfaut [12] obtained a negative performance trade-off between the on-going and prospective-memory tasks; in fact, age differences in prospective memory disappeared when the analysis took into account the performance level in the on-going task. A number of studies were designed where the attentional demands of the on-going activities were manipulated more directly; they provided a mixed pattern of results. In Einstein, McDaniel, Richardson, Guynn, and Cunfer ([16]; Experiment 3) and Otani et al. [33], increasing the attentional demands of the on-going task did not affect prospective memory. In other studies [22,29,34], however, a major influence of the on-going load on prospective memory was observed.

Fortunately, Marsh and Hicks [31] clarified the mixed pattern of findings, by referring to the working memory model of Baddeley [1,2]. No negative impact of the on-going tasks on prospective memory was expected and was observed as long as the on-going tasks involved predominantly the slave systems (i.e., the phonological loop or the visuospatial sketch pad). However, if the on-going tasks implied central executive functioning, they competed with the prospective-memory task in using the same executive resources; in such cases, they obtained a negative effect of the on-going task on prospective memory. In d'Ydewalle, Bouckaert, and Brunfaut [11], the performance in the on-going task involved considerable central executive functioning (solving arithmetic tasks); in agreement with Marsh and Hicks [31], time-based prospective memory among the elderly literally collapsed when the complexity of the on-going task was increased.

We may even go one step further: within the central executive, the different processes need not necessarily compete against each other; they may facilitate each other. Maylor [32] already proposed that compatibility of processing for both the prospective and the on-going tasks is critical. She suggested that whenever different processes are required for the prospective and on-going tasks, constantly shifting from one processing to the other level of processing may impair prospective-memory performance. When there is considerable overlap between the two processing activities necessary to perform each task, no such impairment is predicted. Maylor reported age differences on a prospective event-based task when semantic processing was required on the on-going task and when perceptual processing was required on the prospective task. Mäntylä [30] reported a similar finding for a prospective and on-going task with different processing requirements.

Maylor's task-appropriate processing explanation should in fact predict a better performance when the two tasks share the same level of processing. Our study nicely illustrates such a facilitation: particularly striking is the much better prospective memory in the semantic prospective-memory task when the on-going task requires a semantic analysis than when the on-going task requires perceptual processing. Describing the meaning of the word (the semantic on-going task) prompts the semantic prospective-memory response of pushing a particular button when the name of an animal appears; in such a prospective-memory task, counting the number of letters in the word does not help.

We also obtain a better prospective memory in the perceptual prospective-memory task when the on-going task requires a perceptual analysis than when the on-going task requires semantic processing. However, the difference is much smaller and not significant. The absence of a larger effect among the alcoholics could perhaps be due to a ceiling effect but this explanation does not hold for the Korsakoff patients. In fact, the prospective memory of the Korsakoff patients is surprisingly good in the perceptual prospective-memory task. It should be noted that all participants in the perceptual prospective-memory task (and not in the semantic prospective-memory task) were requested to repeat loudly the instructions of the experimenter before starting the experiment, and this could explain the better performance of the Korsakoff patients in the perceptual prospective-memory task.

As an alternative to the task-appropriate processing explanation, the present findings could also be explained by the instance theory of automaticity [25–27]. The attention hypothesis [28], which is derived from the instance theory, claims that learning is a side effect of attending: people will learn about things they attend to and they will not learn much about the things they do not attend to. As applied to the present experiments, the requested prospective-memory response is a cue-action association which needs to be automatized. Such an automatization of the association will be facilitated when the on-going task directs the attention to what is to be learned for the prospective-memory task. When the two tasks are totally different, the on-going task will not direct the attention to the cue which is relevant for the prospective-memory response.

The task-appropriate processing explanation and the attention hypothesis from the instance theory of automaticity are predicting the same critical interaction between the nature of the on-going and prospective-memory tasks. The attention hypothesis additionally also predicts a gradual strengthening of the prospective association. This improvement of prospective memory indeed was apparent among the three groups in Session 1 of Experiment 1; however, no such improvement was observed in Experiment 2. We therefore slightly favor the task-appropriate processing explanation.

As in Experiment 1, the availability of an external reminder did not help in Experiment 2; apparently, reminders need to be generated internally by the on-going task in order to improve prospective-memory performance.

In the introduction of the present study, we focused on the frontal requirements of the prospective component in prospective-memory tasks, and chose Korsakoff patients because of their frontal lobe deficits. As Korsakoff patients are also amnesic as a result of diencephalic pathology, the retrospective component in the prospective-memory task will also be affected. We therefore predicted a collapse of their prospective-memory functioning. However, contrary to the prediction and the frontal lobe hypothesis in general, their performance in both experiments can surprisingly be good. At Session 2, their prospective memory is not significantly worse than the prospective memory of the alcoholics (Experiments 1 and 2). Even at Session 1, they do show reasonably good performance levels when the prospective-memory and on-going tasks involve the same processing level; in those conditions, their performance also does not differ significantly from the one of the alcoholics (Experiment 2). Following Coeckelbergh [9], we made the requirements of the on-going and prospective-memory tasks as easy as possible, and clearly we succeeded in obtaining a reasonable performance level. The present study, however, does not give an answer whether the remaining performance difference between the Korsakoff patients and the alcoholics reflects frontal or diencephalic deficits, and whether the difference is due to the retrospective or prospective components of the prospective-memory task.

In our preceding work, we already emphasized the importance of detailed analyses of the processing interactions between the prospective-memory and on-going tasks, in order to explain prospective memory; by its nature, a prospective-memory task is always embedded in another task. Our former studies [10–13] focused upon the typically negative trade-off (particularly among older people) between the performance in the on-going task and the prospective-memory task, suggesting that attention to one task is at the expense of performance in the other task. While such a trade-off could not be examined empirically in the present experiments (due to ceiling effects in the on-going tasks), the current study shows that the nature of the on-going task can also positively affect prospective remembering.

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